

University of Essex

MSc Artificial Intelligence

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Unit 9: Collaborative Discussion – Ontology Language Comparison

Initial Discussion

In *Lecture notes in business information processing*, Kalibatiene and Vasilecas assert that “[a]n ontology is a formal, explicit specification of a shared conceptualization” (Kalibatiene, Vasilecas, 2011). The question at hand is to evaluate which of IF, OWL 2, OWL Lite, or RDF is the most useful development platform for software agents running on the Internet.

Of the four options, my view is that OWL 2 is the most appropriate for intelligent agents that are required to operate online or interact with other online intelligent agents. The reasons are straightforward:

1. As a baseline practical matter, OWL 2 is defined by a W3 specification, meaning, its implementation details have been formally structured in accordance with software development best practices (W3.org, 2012).
2. Unlike the other options, OWL 2 offers a variety of robust features that allow for quicker and more accurate development of formally-defined class hierarchies, such as property chains, cardinality restrictions, complex class constructors, disjointness syntax, and granular property characteristics such as symmetry and transitivity (W3.org, 2012).
3. Most importantly, and aligned with Kalibatiene and Vasilecas’ definition of an ontology, OWL 2 requires formal standardization of class characteristics as a prerequisite for using them for logical reasoning, which reduces the chance of errors; the other options (RDF in particular) are more permissive, which may increase the chance of introducing logical reasoning errors (Iglesias-Molina, 2022; Kalibatiene, Vasilecas, 2011; W3.org, 2012).

Finally, it should be noted that this determination should not be taken to mean that OWL 2 will always be the best choice for any given use case; the choice of which ontological development platform to use must rest on the specific fact pattern underlying the use case at hand (Iglesias-Molina, 2022).

References

Iglesias-Molina, A., Cimmino, A., Ruckhaus, E., Chaves-Fraga, D., Raúl García-Castro and Corcho, O. (2022). An ontological approach for representing declarative mapping languages. *Semantic Web*, 15(1), pp.191–221. doi:<https://doi.org/10.3233/sw-223224>.

Kalibatiene, D. and Vasilecas, O. (2011). Survey on Ontology Languages. *Lecture notes in business information processing*, pp.124–141. doi:https://doi.org/10.1007/978-3-642-24511-4_10.

W3.org. (2012). *OWL 2 Web Ontology Language Document Overview (Second Edition)*. [online] Available at: <https://www.w3.org/TR/owl2-overview/>.